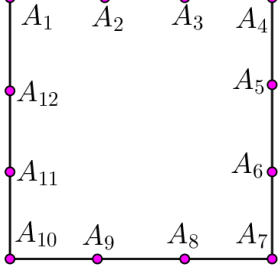


	نموذج لمسابقة في مادة الرياضيات عدد المسائل: خمسة الأستاذ محمد دحبول: 03/334396 الأستاذ: منير حمية: 03/376654 Jazwat Elrisala علم وخبر رقم : 1629	لبنان  الغد
المدة: أربع ساعات	اللغة: الفرنسية	الصف: العلوم العامة

Question 1:

Pour chaque question, une seule réponse est correcte ; trouver la bonne réponse et justifiez votre choix.

N°	Questions	Réponses		
		a	b	c
1.	Le nombre de droites Passant par ces 12 Points est 	44	92	46
2.	Soit $f(x) = x - \frac{2e^x}{1+e^x}$, $x \in \mathbb{R}$, donc la pente, dans un système orthonormé, de la droite tangente en tout point de (C_f) est toujours	Plus grande que $\frac{1}{2}$	égal à $\frac{1}{2}$	Plus petite que $\frac{1}{2}$
3.	soit $f(x) = x - 1 - \frac{2e^x + 3}{1+e^x}$, $x \in \mathbb{R}$, donc L'équation réduite de l'asymptote oblique de (C_f) en $+\infty$ est	$y = x - 1$	$y = -x + 1$	$y = x - 3$
4.	Let $f(x) = \frac{e^{2x} - 3e^x + 2}{1+e^x}$, $x \in \mathbb{R}$, donc La courbe (C_f), dans un système orthonormé, se trouve au dessous de l'axe des abscisses lorsque	$x > \ln 2$	$0 < x < \ln 2$	$x < 0$

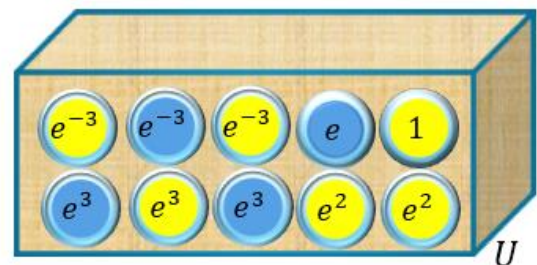
Question 2 :

Une urne U contient dix boules indiscernables au toucher,
Portant les numéros $1; e; e^2; e^2; e^3; e^3; e^3; e^{-3}; e^{-3}; e^{-3}$.

On tire successivement, sans remise, deux boules de U .

Notons "a" le nombre figurant sur la première boule tirée et
par "b" le numéro figurant sur la deuxième boule tirée.

On considère les événements suivants :



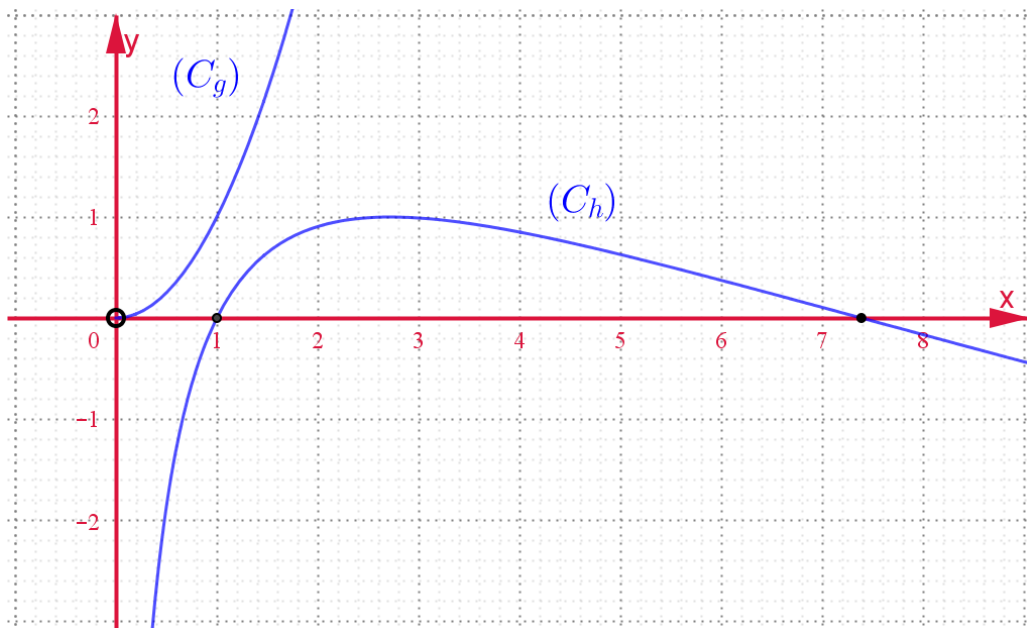
A: " $\ln(ab)=0$ "; B: " $\ln(a) \times \ln(b)=0$ "; C: " $\frac{\ln(a)}{\ln(b)}=0$ "; D: " $\ln(b) < 0$ ".

- 1) **a)** Montrer que la probabilité de l'événement **A** est égale à $\frac{1}{5}$.
b) Démontrez que les deux événements **A** et **B** sont équiprobables.
- 2) **a)** Calculer la probabilité de l'événement **C** et montrer que la probabilité de **D** est égale à $3/10$.
b) Connaissant que $\ln(b) < 0$. Quelle est la probabilité que l'événement **C** est réaliser ?
c) Calculer la probabilité conditionnelle $P(C/B)$.
- 3) Calculer la probabilité que l'évènement **C** soit réalisé, sachant que l'un des évènements **B** ou **D** est réalisé.
- 4) On désigne par **X** la variable aléatoire algébrique égale à $\ln a + \ln b$.
a) Vérifier que $P(X = 4) = 4/45$.
b) Déterminer les onze valeurs de **X**.
c) Calculer la probabilité de l'évènement **E** : " $X \leq 4$ ".

Question 3 :

Partie A

La figure ci-dessous représente les courbes (C_g) et (C_h) de g et h , chacune définie sur $]0; +\infty[$, respectivement par : $g(x) = x^2$ et $h(x) = 2\ln x - (\ln x)^2$.



- 1) **a)** Justifier graphiquement que, pour tout $x \in]0; +\infty[$, on a $g(x) - h(x) > 0$.
b) Deducire que pour tout $x \in]0; +\infty[$, on a $\frac{2\ln x - (\ln x)^2}{x^2} < 1$.
- 2) **a)** Vérifier que la fonction $K: x \rightarrow x\ln x - x$ est une primitive $]0; +\infty[$ de la fonction $x \rightarrow \ln x$.
b) deduire que $\int_1^{e^2} \ln x \cdot dx = 1 + e^2$.
c) À l'aide de l'intégration par parties, montrez que $\int_1^{e^2} (\ln x)^2 \cdot dx = 2e^2 - 2$.
d) Résoudre sur l'intervalle $]0; +\infty[$, l'équation $h(x) = 0$, et en déduire les coordonnées du point d'intersection entre (C_h) et $(x'x)$.
e) Calculez, en unité carré, l'aire de la région limitée par (C_h) and $(x'x)$.

Partie B

Considérons la fonction numérique f définie sur $]0; +\infty[$ par :

$$f(x) = x - \frac{(\ln x)^2}{x}$$

On désigne par (C_f) la courbe représentative de la fonction f dans un système orthonormé $(O; \vec{i}; \vec{j})$.

1) a) Vérifier que $\lim_{x \rightarrow 0^+} f(x) = -\infty$. Interpréter graphiquement.

b) Vérifier $\lim_{x \rightarrow +\infty} \frac{(\ln x)^2}{x} = 0$ et calculer $\lim_{x \rightarrow +\infty} f(x)$

c) En déduire que la droite $(d): y = x$ une asymptote oblique de (C_f) quand x tend vers $+\infty$.

d) Montrer que (d) est tangente à (C_f) en $A(1; 1)$.

2) a) Montre que pour tout $x \in]0; +\infty[$, $f'(x) = 1 - \frac{2\ln x - (\ln x)^2}{x^2}$.

b) Justifier que f est strictement croissante $]0; +\infty[$ et dresser son tableau de variation.

c) Demontrez qu'il existe un seul point B de (C_f) dont la tangente (T) est strictement parallèle à (d) .

d) Montre que l'équation $f(x) = 0$ a une racine unique β . Vérifier que $\beta \in]0.5; 0.6[$ et que $\ln(\beta) = -\beta$.

e) Construire dans $(O; \vec{i}; \vec{j})$, les droites (d) , (T) et la courbe (C_f) .

3) a) Montre que la fonction f admet sur $]0; +\infty[$ une fonction réciproque f^{-1} .

b) Déterminer le domaine de définition de f^{-1} .

c) Montrer que f^{-1} est dérivable en **zero** et que $(f^{-1})'(0) = \frac{\beta}{2 + 2\beta}$.

d) Tracer dans le même système la courbe $(C_{f^{-1}})$.

4) Soit B' le point symétrique de B par rapport à (d) et E l'intersection de (d) avec la droite passant par B et perpendiculaire à $(x'x)$.

a) Montrez que le triangle BEB' est un triangle rectangle isocèle de sommet E .

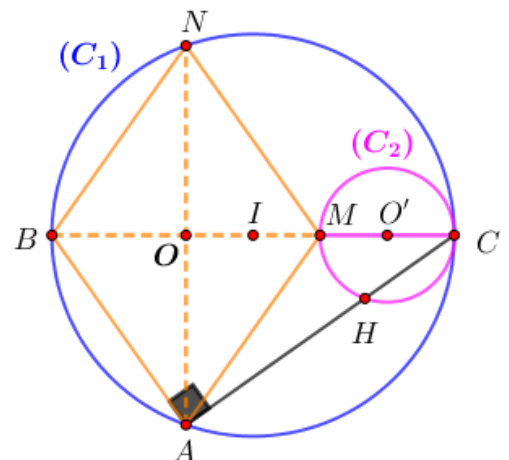
b) Vérifiez que l'aire du triangle BEB' est égale à $8 \times e^{-4}$ (unité)².

b) Calculez, en carrés d'unité, l'aire de la région limitée par les courbes (C_f) and $(C_{f^{-1}})$ et le segment $[BB']$.

Question 4

D'après la figure ci-contre, considérons les informations suivantes :

- (C_1) est le cercle de centre I et de diamètre $[BC]$, avec $BC = 6 \text{ cm}$.
- M est un point du segment $[BC]$ tel que $BC = 3 \times MC$.
- Le point I est le milieu du segment $[BM]$.
- (C_2) est le cercle de centre O' et de diamètre $[CM]$.
- A et N sont deux points de (C_1) tels que : $(\overrightarrow{AC}; \overrightarrow{AB}) = \pi/2 [2\pi]$ et $AMNB$ est un losange.
- (AC) coupe (C_2) au point H ($H \neq C$).



- 1) **a)** Démontrer que les deux droites (HM) et (AB) sont parallèles.
b) En déduire que les trois points H, M et N sont alignés.
c) Montre que $AB = 3 \times HM$ et $HA^2 = AB^2 - HM^2$.
- 2) Désignez par S la similitude plane directe du centre H qui transforme A en M .
a) Précisez l'angle α de S et prouvez que son rapport $K = \sqrt{2}/4$.
b) Déterminer les images des deux droites (AO) et (MH) par S . En déduire $S(N)$.
- 3) Montrer que $S(O) = O'$ et en déduire que (HO) est tangente à (C_2) en H .
- 4) Soit (C'_2) l'image de (C_2) sous S . Prouvez que $(O'H)$ est la tangente au cercle (C'_2) en H .
- 5) Le plan est qualifié de système orthonormé $(O; \vec{u}; \vec{v})$ avec $\vec{u} = \overrightarrow{OI}$
a) Ecrire la forme complexe de S .
b) Déduisez les coordonnées du point H .
c) Démontrer que le point D symétrique de M par rapport à H appartient à (C_1)

Question 5 :

Le plan complexe (P) est rapporté à un système orthonormé direct $(O; \vec{u}; \vec{v})$.

Soient A, B et C les points d'affixes respectifs $i, 1 + i$ et $-1 + i$.

À chaque point M du plan (P) d'affixe $z \neq i$, on associe le point M' du plan (P) d'affixe z' de sorte que

$$z' = \frac{iz + 2}{z - i}.$$

- 1) Montrer que si $z \neq i$ alors $(z - i)(z' - i) = 1$.
- 2) Déterminez l'ensemble des points M dans le plan (P) tels que
- 3) **a)** Montrer que les points A, M et M' sont alignés si et seulement si $(z - i)^2$ est réel.
b) Déduisez l'ensemble des points M dans le plan (P) tels que A, M et M' soient colinéaires
- 4) Montrer que pour tout M d'affixe $z \neq i$, $AM \times AM' = 1$ et $(\overrightarrow{AB}; \overrightarrow{AM}) = (\overrightarrow{AM'}; \overrightarrow{AB}) [2\pi]$.

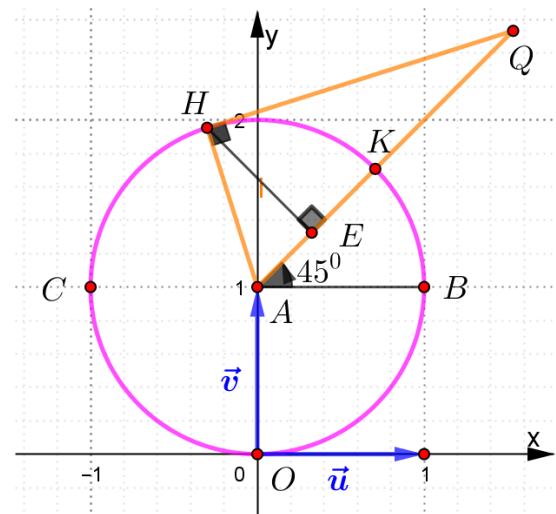
D'après la figure ci-contre, tenez compte des informations suivantes :

** (C) est un cercle de centre A et de rayon 1.

** K et Q sont les points ayant les affixes $z_K = i + e^{i\frac{\pi}{4}}$
et $z_Q = i + re^{i\frac{\pi}{4}}, r > 1$.

** H est un point de (C) tel que AHQ est triangle rectangle en H et E est le projeté orthogonal de H sur la droite (AK) .

- 5) Soit K' le point d'affixe z'_K
a) Calculer la distance AK' et donner une mesure de $(\overrightarrow{AB}; \overrightarrow{AK'})$
b) Placer le point K' sur la figure donnée.
c) Déterminer l'ensemble des points M' quand M décrit la demi droite $[AK)$ privé du point A
- 6) **a)** Montrer que $AQ \times AE = 1$.
b) Construire le point Q' d'affixe z'_Q .



Wishing you all the best

Correction of the exam 2026. L.S.

Question 1:

- 1) The number of straight Lines passing through These 12 points is

$$N = C_{12}^2 - 4 \times C_4^2 + 1 + 1 + 1 + 1 = 66 - 4 \times 6 + 4$$

Then $N = 46$ and therefore, "c" is true.

- 2) $f(x) = x - \frac{2e^x}{1+e^x}, x \in \mathbb{R}$.

f is differentiable over $\mathbb{R}$ and for all real numbers,

$$f'(x) = 1 - 2 \left(\frac{e^x(e^x + 1) - e^x \times e^x}{(1 + e^x)^2} \right) = \frac{1 + 2e^x + e^{2x} - 2e^x}{(1 + e^x)^2} = \frac{1 + e^{2x}}{(1 + e^x)^2}$$

$$f'(x) - \frac{1}{2} = \frac{1 + e^{2x}}{(1 + e^x)^2} - \frac{1}{2} = \frac{2 + 2e^{2x} - 1 - 2e^x - e^{2x}}{2(1 + e^x)^2} = \frac{1 - e^x + e^{2x}}{2(1 + e^x)^2} = \frac{(1 - e^x)^2}{2(1 + e^x)^2} \geq 0$$

Therefore, the slope, in an orthonormal system, of the tangent line at any point of (C_f) is always greater than 0.5. therefore, "a" is true.

- 3) $f(x) = x - 1 - \frac{2e^x + 3}{1 + e^x}, x \in \mathbb{R}$

$$\lim_{x \rightarrow +\infty} [f(x) - (x - 3)] = \lim_{x \rightarrow +\infty} \left(2 - \frac{2e^x + 3}{1 + e^x} \right) = 2 - 2 = 0.$$

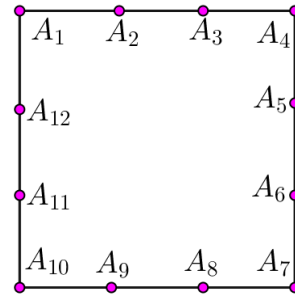
Then, the reduced equation of the oblique asymptote to (C_f) at $+\infty$ is $y = x - 3$.

Therefore, "c" is true.

- 4) Let $f(x) = \frac{e^{2x} - 3e^x + 2}{1 + e^x}, x \in \mathbb{R}$, then,

The curve (C_f) , in an orthonormal system, is below the axis of abscissas when

$$e^{2x} - 3e^x + 2 < 0 \Rightarrow (e^x - 1)(e^x - 2) < 0 \Rightarrow 1 < e^x < 2 \Rightarrow 0 < x < \ln 2. \text{ Therefore, "b" is true.}$$



Question 2:

Initial Data Analysis & Configuration

An urn U contains 10 balls which are identical and indistinguishable to the touch. We draw two balls sequentially and without replacement. Let 'a' be the number on the first ball drawn, and 'b' be the number on the second ball drawn. The distribution of the values and their respective natural logarithms is detailed below:

Ball Value (x)	Quantity in Urn	Natural Logarithm $\ln(x)$
1	1	$\ln(1) = 0$
e	1	$\ln(e) = 1$
e^2	2	$\ln(e^2) = 2$
e^3	3	$\ln(e^3) = 3$
e^{-3}	3	$\ln(e^{-3}) = -3$

Total Number of Outcomes (Sample Space Ω):

Since the drawing is successive and without replacement, the total number of ordered pairs is:

$$\text{Card}(\Omega) = 10 \times 9 = 90 \text{ outcomes.}$$

1)

a) Show that the probability of event A is equal to $1/5$.

Event A is defined as: ' $\ln(ab) = 0$ '. According to logarithmic identities, $\ln(ab) = \ln(a) + \ln(b) = 0$, which yields $\ln(b) = -\ln(a)$. Let's determine the satisfying ordered pairs:

• **Case 1: One log value is 3 and the other is -3 .**

- Drawing e^3 first (3 choices), then e^{-3} (3 choices) $\rightarrow 3 \times 3 = 9$ outcomes.

- Drawing e^{-3} first (3 choices), then e^3 (3 choices) $\rightarrow 3 \times 3 = 9$ outcomes.

• **Case 2: Both log values are 0 ($\ln(a) = 0$ and $\ln(b) = 0$).**

- This requires drawing the ball '1' twice. Since the process is without replacement and there is only one ball '1' in the urn, this case is impossible (0 outcomes).

$$\text{Total favorable outcomes} = 9 + 9 = 18$$

$$P(A) = 18/90 = 1/5$$

b) Prove that events A and B are equiprobable.

Event B is defined as: ' $\ln(a) \times \ln(b) = 0$ '. A product equals zero if and only if at least one factor is zero. This implies $\ln(a) = 0$ or $\ln(b) = 0$ (meaning at least one of the selected balls must be '1'):

• **Case 1: The first ball is '1' ($\ln(a) = 0$) and the second is any remaining ball.**

- Choices: $1 \times 9 = 9$ outcomes.

• **Case 2: The first ball is any other ball and the second ball is '1' ($\ln(b) = 0$).**

- Choices: $9 \times 1 = 9$ outcomes.

$$\text{Total favorable outcomes} = 9 + 9 = 18$$

$$P(B) = 18/90 = 1/5$$

Since $P(A) = P(B) = 1/5$, both events are verified to be equiprobable.

2)

a) Calculate $P(C)$ and show that $P(D) = 3/10$.

• **Probability of Event C ($\ln(a) / \ln(b) = 0$):**

A quotient is zero if and only if its numerator is zero and its denominator is non-zero. Hence, we require $\ln(a) = 0$ AND $\ln(b) \neq 0$.

- The first ball must be '1' (1 choice).

- The second ball must be any ball except '1' (9 choices).

- Favorable outcomes = $1 \times 9 = 9$.

$$P(C) = 9/90 = 1/10.$$

• **Probability of Event D ($\ln(b) < 0$):**

This implies that the second ball drawn must have a negative logarithm. The only eligible candidates are the 3 balls labeled e^{-3} . Let's count via sequential combinations:

- Sub-case D1: First ball is an e^{-3} ball (3 choices), second ball is another e^{-3} ball (2 choices) $\rightarrow 3 \times 2 = 6$ outcomes.

- Sub-case D2: First ball is NOT an e^{-3} ball (7 choices), second ball is an e^{-3} ball (3 choices) $\rightarrow 7 \times 3 = 21$

outcomes.

- Total favorable outcomes = $6 + 21 = 27$.

$$P(D) = 27/90 = 3/10.$$

b) Given that $\ln(b) < 0$, what is the probability that event C is realized?

This requires evaluating the conditional probability $P(C|D) = P(C \cap D)/P(D)$.

The intersection event $C \cap D$ translates to: $\ln(a) = 0$ AND $\ln(b) < 0$.

- First ball must be '1' (1 choice).

- Second ball must be one of the e^{-3} balls (3 choices).

- Favorable outcomes for $C \cap D = 1 \times 3 = 3$.

$$P(C \cap D) = 3/90 = 1/30.$$

$$P(C|D) = P(C \cap D)/P(D) = (3/90)/(27/90) = 3/27 = 1/9.$$

c) Calculate $P(C|B)$.

We have, $P(C|B) = P(C \cap B)/P(B)$. Let's inspect the logical relationship between C and B :

Event C implies that $\ln(a) = 0$ and $\ln(b) \neq 0$. If C occurs, the product $\ln(a) \times \ln(b) = 0 \times \ln(b) = 0$, which directly satisfies event B . Thus, event C is a subset of event B ($C \subset B$), meaning $C \cap B = C$.

$$P(C|B) = P(C)/P(B) = (1/10)/(1/5) = 5/10 = 1/2.$$

3) Calculer la probabilité que l'évènement C soit réalisé, sachant que l'un des évènements B ou D est réalisé.

$$* P(C|(B \cup D)) = \frac{P(C \cap (B \cup D))}{P(B \cup D)} = \frac{P((C \cap B) \cup (C \cap D))}{P(B \cup D)} = \frac{P(C \cap B) + P(C \cap D) - P(C \cap B \cap D)}{P(B) + P(D) - P(B \cap D)}$$

* We Have calculated : $P(B) = 1/5 = 18/90$, $P(D) = 3/10 = 27/90$, $P(B \cap D) = 3/90$

* $(C \cap D) \Rightarrow (\ln a = 0 \text{ AND } \ln b < 0) \Rightarrow \ln a \times \ln b = 0 \Rightarrow C \cap D \subset B \Rightarrow C \cap D \cap B = C \cap D$

$$P(C \cap D) = P(C \cap B \cap D) \Rightarrow P(C \cap D) - P(C \cap B \cap D) = 0 \Rightarrow$$

$$P(C|(B \cup D)) = \frac{P(C \cap B)}{P(B) + P(D) - P(B \cap D)} = \frac{\frac{9}{90}}{\frac{42}{90}} = \frac{9}{42} = \frac{3}{14}$$

4) Let X be the algebraic random variable equal to $\ln(a) + \ln(b)$.

a) Verify that $P(X = 4) = 4/45$.

To attain a sum of $X = 4$, we evaluate the valid arrangements of $\ln$ values:

• **Case 1: $\ln(a) = 1$ and $\ln(b) = 3$ (ball ' e ' then ball ' e^3 ')**

- Outcomes: $1 \times 3 = 3$.

• **Case 2: $\ln(a) = 3$ and $\ln(b) = 1$ (ball ' e^3 ' then ball ' e ')**

- Outcomes: $3 \times 1 = 3$.

• **Case 3: $\ln(a) = 2$ and $\ln(b) = 2$ (ball ' e^2 ' then another ball ' e^2 ')**

- Outcomes: $2 \times 1 = 2$ (since there are two e^2 balls, drawing without replacement is possible).

$$\text{Total outcomes for } (X = 4) = 3 + 3 + 2 = 8$$

$$P(X = 4) = 8/90 = 4/45$$

The targeted value is verified.

b) Determine the eleven values of X .

The possible scores for X are generated by compiling all permutations of two valid $\ln$ values from the available set $\{-3, 0, 1, 2, 3\}$ while adhering to structural availability constraints:

$$1. (-3) + (-3) = -6 \quad (\text{Valid: 3 balls of } e^{-3} \text{ exist})$$

$$2. (-3) + 0 = -3$$

$$3. (-3) + 1 = -2$$

$$4. (-3) + 2 = -1$$

$$5. (-3) + 3 = 0$$

$$6. 0 + 1 = 1$$

$$7. 0 + 2 = 2$$

$$8. 0 + 3 = 3 \quad (\text{or } 1 + 2 = 3)$$

$$9. 1 + 3 = 4 \quad (\text{or } 2 + 2 = 4)$$

$$10. 2 + 3 = 5$$

$$11. 3 + 3 = 6 \quad (\text{Valid: 3 balls of } e^3 \text{ exist})$$

Hence, the set of the eleven possible values of X is:

$$X(\Omega) = \{-6, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6\}$$

c) Calculer la probabilité de l'évènement $E : "X \leq 4"$.

$$P(X \leq 4) = 1 - P(X = 5) + P(X = 6) = 1 - \frac{2}{15} + \frac{1}{15} = \frac{12}{15} = \frac{4}{5}$$

Question 3:**Part A**

1) a) According to the figure, (C_g) is above (C_h) then, For all $x \in]0; +\infty[$, $g(x) - h(x) > 0$.

b) Let $x \in]0; +\infty[$, then $g(x) - h(x) > 0$ this implies that $x^2 > 2\ln x - (\ln x)^2$.

Divide by the strictly positive real number x^2 we get'

$$\frac{x^2}{x^2} > \frac{2\ln x - (\ln x)^2}{x^2}, \text{ this leads to } \frac{2\ln x - (\ln x)^2}{x^2} < 1.$$

2) a) The function K is differentiable over $]0; +\infty[$. For $x \in]0; +\infty[$, $K(x) = x\ln x - x$ then,

$$K'(x) = 1 \times \ln x + \frac{x}{x} - 1 = \ln x + 1 - 1 = \ln x.$$

Therefore, K is a primitive on $]0; +\infty[$ of the function $x \rightarrow \ln x$.

$$b) \int_1^{e^2} \ln x \cdot dx = (x\ln x - x) \Big|_1^{e^2} = e^2 \ln e^2 - e^2 - 0 + 1 = 2e^2 - e^2 + 1 = e^2 + 1.$$

$$c) \int_1^{e^2} (\ln x)^2 \cdot dx = \int_1^{e^2} (x)' (\ln x)^2 \cdot dx = x(\ln x)^2 \Big|_1^{e^2} - \int_1^{e^2} \frac{2x\ln x}{x} \cdot dx = e^2 (\ln e^2)^2 - 0 - \int_1^{e^2} 2\ln x \cdot dx$$

$$\int_1^{e^2} (\ln x)^2 \cdot dx = 4e^2 - 2(e^2 + 1) = 2e^2 - 2.$$

d) Let $x \in]0; +\infty[$. $h(x) = 0 \Leftrightarrow 2\ln x - (\ln x)^2 = 0 \Leftrightarrow \ln x(2 - \ln x) = 0 \Leftrightarrow \ln x = 0$ or $\ln x = 2$.
 $h(x) = 0 \Leftrightarrow x = 1$ or $x = e^2$.

Therefore, the two curves (C_h) and $(x'x)$ meet at the two points of coordinates $(1; 0)$ and $(e^2; 0)$.

e) The area of the region bounded between (C_h) and $(x'x)$ is equal to

$$\left(\int_1^{e^2} h(x) \cdot dx \right) \times (\text{unit})^2 \text{ because for all } x \in [1; e^2], h(x) \geq 0.$$

$$\int_1^{e^2} h(x) \cdot dx = \int_1^{e^2} [2\ln x - (\ln x)^2] dx = 2(e^2 + 1) - (2e^2 - 2) = 4.$$

Therefore, the required area is equal to $4 \times (\text{unit})^2$

Part B

$$x \in]0; +\infty[, f(x) = x - \frac{(\ln x)^2}{x}$$

$$1) a) \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \left(x - \frac{(\ln x)^2}{x} \right) = -\frac{\infty}{0^+} = (-\infty) \left(\frac{1}{0^+} \right) = (-\infty)(+\infty) = -\infty.$$

Then, the straight line $(y'y): x = 0$ is a vertical asymptote to the curve (C_f) .

$$b) \lim_{x \rightarrow +\infty} \frac{(\ln x)^2}{x} = \frac{\infty}{\infty}. \text{ Using H. R. we get, } \lim_{x \rightarrow +\infty} \frac{(\ln x)^2}{x} = \lim_{x \rightarrow +\infty} \frac{\frac{2\ln x}{x}}{1} = \lim_{x \rightarrow +\infty} \frac{2\ln x}{x} = 0.$$

$$\text{Calculate } \lim_{x \rightarrow +\infty} f(x) = \lim_{x \rightarrow +\infty} \left(x - \frac{(\ln x)^2}{x} \right) = +\infty - 0 = +\infty.$$

$$c) \lim_{x \rightarrow +\infty} (f(x) - x) = \lim_{x \rightarrow +\infty} \left(x - \frac{(\ln x)^2}{x} - x \right) = \lim_{x \rightarrow +\infty} \frac{(\ln x)^2}{x} = 0.$$

Then, the straight line $(d): y = x$ is an oblique asymptote of (C_f) in the $N(+\infty)$.

Now, $M(x; y) \in (C_f) \cap (d) \Leftrightarrow M(x; y) \in (C_f)$ and $M(x; y) \in (d) \Leftrightarrow y = x - \frac{(\ln x)^2}{x}$ and $y = x$

$$x - \frac{(\ln x)^2}{x} = x \Leftrightarrow \frac{(\ln x)^2}{x} = 0 \Leftrightarrow (\ln x)^2 = 0 \Leftrightarrow x = 1 \text{ is a double root. For } x = 1, y = 1$$

Therefore, the straight line (d) is tangent to (C_f) at the point $A(1; 1)$.

2) a) The function f is differentiable over its domain of definition $]0; +\infty[$. And for all $x \in]0; +\infty[$,

$$f'(x) = 1 - \frac{x \left(\frac{2\ln x}{x} \right) - 1 \times (\ln x)^2}{x^2} = 1 - \frac{2\ln x - (\ln x)^2}{x^2}.$$

b) According to **Part A. 1) b)**: we have, for all real number $x \in]0; +\infty[$,

$$\frac{2\ln x - (\ln x)^2}{x^2} < 1 \Rightarrow 1 - \frac{2\ln x - (\ln x)^2}{x^2} > 0 \Rightarrow f'(x) > 0.$$

Therefore, f is strictly increasing over $]0; +\infty[$.

table of variation of the function f over $]0; +\infty[$:

x	0	$+\infty$
$f'(x)$		+
$f(x)$	$-\infty$	$+\infty$

c) Let $M(x; y) \in (C_f)$ such that the tangent line to (C_f) is strictly parallel to (d) . This leads us to say that the slope of strictly parallel to (T) is equal to the slope of strictly parallel to (d) .

$$f'(x) = 1 \text{ and } x \in]0; +\infty[- \{1\} \Leftrightarrow 1 - \frac{2\ln x - (\ln x)^2}{x^2} = 1 \Leftrightarrow \frac{2\ln x - (\ln x)^2}{x^2} = 0 \Leftrightarrow$$

$$2\ln x - (\ln x)^2 = 0 \Leftrightarrow \ln x(2 - \ln x) = 0 \Leftrightarrow \ln x = 0 \text{ or } \ln x = 2 \Leftrightarrow x = 1 (\text{rejected}) \text{ or } x = e^2.$$

$$x = e^2, y = e^2 - \frac{(\ln e^2)^2}{e^2} = e^2 - \frac{4}{e^2} = e^2 - 4e^{-2}.$$

Therefore, there exists only one point $B(e^2; e^2 - 4e^{-2})$ of (C_f) whose tangent line (T) is strictly parallel to (d) .

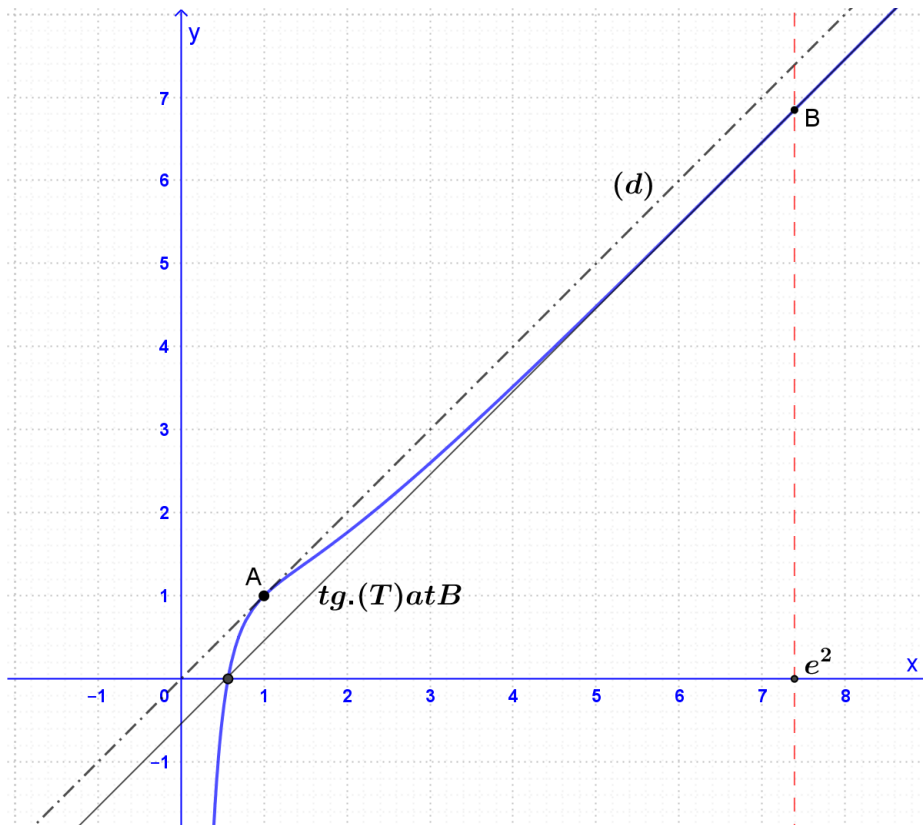
d) According to the table of variation of f : f is continuous, strictly increasing over $]0; +\infty[$ and it goes from negative to positive, this leads that the equation $f(x) = 0$ has only one root $\beta \in]0; +\infty[$.

0.5 and 0.6 $\in]0; +\infty[$ and $f(0.5) \times f(0.6) < 0$, then $\beta \in]0.5; 0.6[$.

$$f(\beta) = 0 \Leftrightarrow \beta - \frac{(\ln \beta)^2}{\beta} = 0 \Leftrightarrow (\ln \beta)^2 = \beta^2 \Leftrightarrow \ln \beta = \beta \text{ or } \ln \beta = -\beta.$$

But $\ln \beta < 0$ because $\beta \in]0.5; 0.6[$. Thus $\ln(\beta) = -\beta$.

e) Plot in $(O; \vec{i}; \vec{j})$ (d) , (T) and the curve (C_f) .



3) a) The ft. f , on $]0; +\infty[$, is continuous and strictly increasing then, it has an inverse function f^{-1} .

b) The domain f^{-1} of is defined by $D_{f^{-1}} = f(]0; +\infty[) =]\lim_{x \rightarrow 0^+} f(x); \lim_{x \rightarrow +\infty} f(x)[$

Thus, $D_{f^{-1}} =]-\infty; +\infty[= \mathbb{R}$.

c) Prove that f^{-1} is differentiable at zero and that $(f^{-1})'(0) = \frac{\beta}{2 + 2\beta}$

Recall:

$(x \in D_{f^{-1}} = \mathbb{R} \text{ and } y = f^{-1}(x)) \Leftrightarrow (x \in D_f =]0; +\infty[\text{ and } x = f(y))$

$x = f(y)$. Differentiate with respect to x , we get:

$1 = y' \times f'(y)$, then $y' = \frac{1}{f'(y)}$, then $(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$ with $f^{-1}(x) \neq 0$

Property: if f is differentiable at a and $f'(a) \neq 0$ then, f^{-1} is differentiable at $b = f(a)$ and

$(f^{-1})'(f(a)) = \frac{1}{f'(a)}$. Or $(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}$.

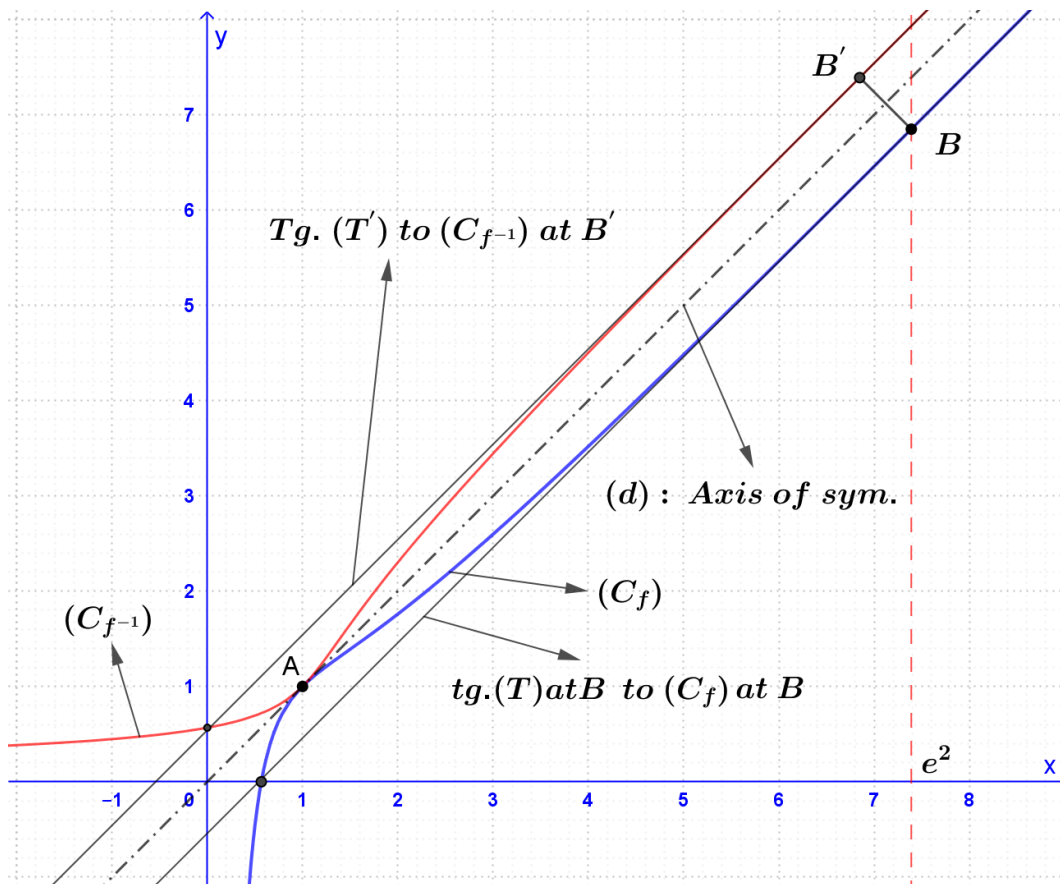
Now, f is differentiable at β and $f'(\beta) \neq 0$ then, f^{-1} is differentiable at $f(\beta) = 0$ and

$(f^{-1})'(0) = \frac{1}{f'(f^{-1}(0))} = \frac{1}{f'(\beta)}$.

$f'(\beta) = 1 - \frac{2\ln\beta - (\ln\beta)^2}{\beta^2} = 1 - \frac{-2\beta - (-\beta)^2}{\beta^2} = \frac{\beta^2 + 2\beta + (-\beta)^2}{\beta^2} = \frac{2\beta^2 + 2\beta}{\beta^2} = \frac{2 + 2\beta}{\beta}$

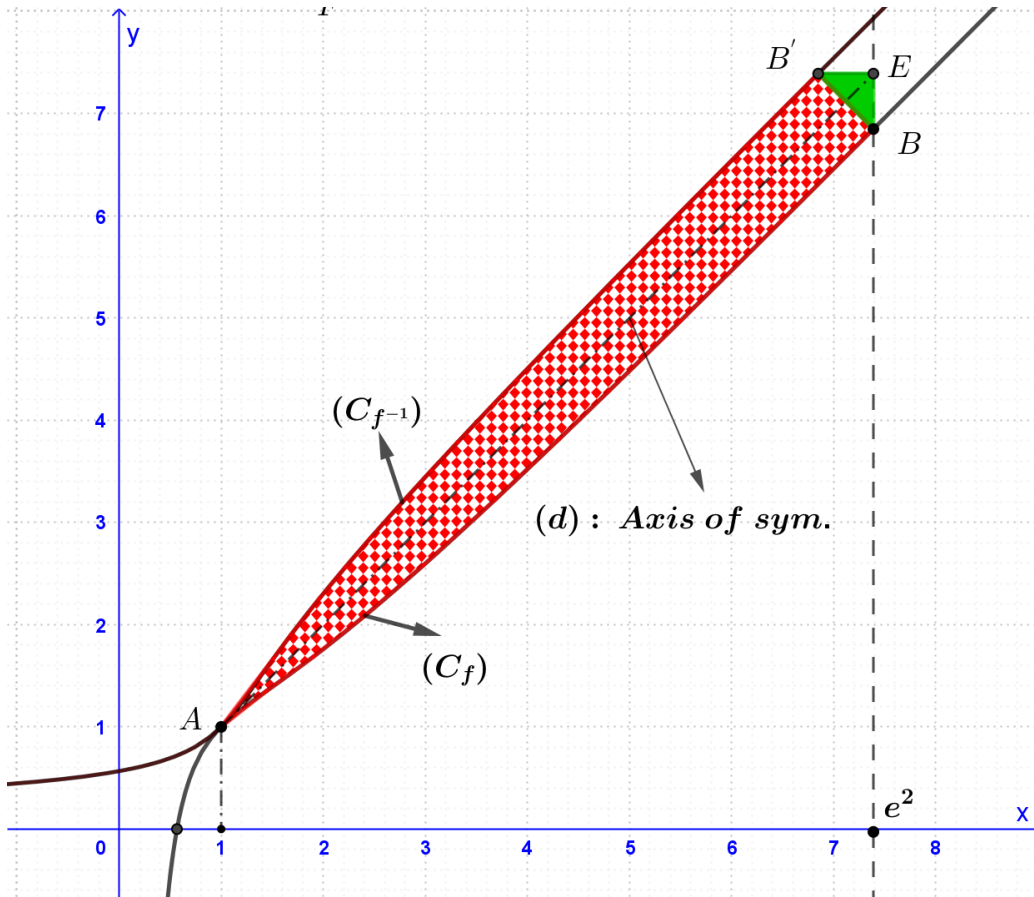
Then, $\frac{1}{f'(\beta)} = \frac{\beta}{2 + 2\beta}$. Therefore, $(f^{-1})'(0) = \frac{\beta}{2 + 2\beta}$

d) Plot in the same system the curve $(C_{f^{-1}})$.



- The two curves (C_f) and $(C_{f^{-1}})$ are symmetric with respect to the straight line $(d): y = x$.
- (d) is the oblique asymptote to (C_f) in the $N(+\infty)$ then, (d) is the oblique asymptote to $(C_{f^{-1}})$ in the $N(+\infty)$
- (d) is tangent to (C_f) at $A(1; 1)$, then (d) is tangent to $(C_{f^{-1}})$ at $A' \equiv A(1; 1)$.
- (T) is parallel to (d) and tangent to (C_f) at $B(e^2; e^2 - 4e^{-2})$, then (T') is parallel to (d) and tangent to $(C_{f^{-1}})$ at $B'(e^2 - 4e^{-2}; e^2)$.

4)



a) We have, $E(e^2; e^2) \in (d): y = x; B(e^2; e^2 - 4e^{-2}) \in (C_f); B'(e^2 - 4e^{-2}; e^2)$

$$\overrightarrow{EB}(0; -4e^{-2}); \overrightarrow{EB'}(-4e^{-2}; 0).$$

$$\overrightarrow{EB} \cdot \overrightarrow{EB'} = 0 = 0 = 0 \text{ and } \|\overrightarrow{EB}\| = 4e^{-2} = \|\overrightarrow{EB'}\|$$

Therefore, the triangle BEB' is right isosceles at E .

b) Area of the triangle BEB' is equal to $4e^{-2} \times 4e^{-2} \times 0.5 \text{ (unit)}^2 = 8 \times e^{-4} \times \text{(unit)}^2$

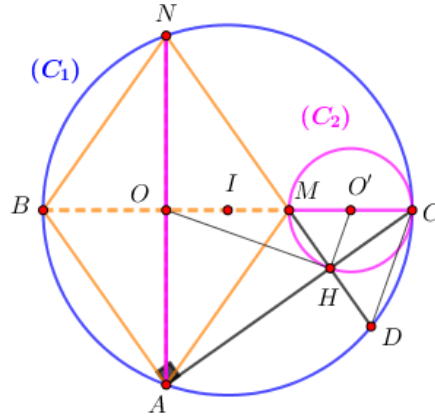
Therefore, the area of the triangle BEB' is equal to $8 \times e^{-4} \times \text{(unit)}^2$.

b) Let S be the area of the region bounded between the two curves (C_f) and $(C_{f^{-1}})$ and the segment $[BB']$. Then,

$$S = \left(2 \int_1^{e^2} (y_{(a)} - f(x)) dx - 8 \times e^{-4} \right) \times \text{(unit)}^2 = \left(\frac{16}{3} - \frac{8}{e^4} \right) \times \text{(unit)}^2, \text{ because,}$$

$$2 \int_1^{e^2} (y_{(d)} - f(x)) dx = 2 \int_1^{e^2} (y_{(d)} - f(x)) dx = 2 \int_1^{e^2} \frac{(\ln x)^2}{x} dx = 2 \int_1^{e^2} (\ln x)' (\ln x)^2 dx = \frac{16}{3}.$$

Question 4:



1) Geometric Properties

a) Show that the two straight lines (HM) and (AB) are parallel.

- Angle $\angle BAC$: The point A lies on the circle (C_1) with diameter $[BC]$. An angle inscribed in a semicircle is a right angle, so $\angle BAC = 90^\circ$. This implies that $(AB) \perp (AC)$.
- Angle $\angle MHC$: The point H lies on the circle (C_2) with diameter $[CM]$. Similarly, the inscribed angle facing its diameter is a right angle, so $\angle MHC = 90^\circ$. This implies that $(HM) \perp (AC)$.
- **Conclusion:** Since both straight lines (AB) and (HM) are perpendicular to the same straight line (AC) , they must be parallel to each other: $(HM) \parallel (AB)$.

b) Deduce that the three points H , M , and N are collinear.

- Since $AMNB$ is a rhombus, its opposite sides are parallel: $(NM) \parallel (AB)$
- From part 1) a, we have already proven that: $(HM) \parallel (AB)$
- According to Euclid's parallel postulate, through a given point M not on the line (AB) , there exists only one unique line parallel to (AB) .
- Since both lines (NM) and (HM) pass through M and are parallel to (AB) , they must be the exact same straight line. Thus, the points H , M , and N are collinear.

c) Show that $AB = 3 \times HM$ and $HA^2 = AB^2 - HM^2$.

- **Proving $AB = 3 \times HM$:**

In triangle CAB , the line (HM) is parallel to (AB) . Applying Thales's Theorem (or triangle similarity $\triangle CHM \sim \triangle CAB$), (congruent triangles) we get the side ratios:

$$HM/AB = CM/CB$$

We are given that $BC = 3 \times MC$, which means $CM/CB = 1/3$. Substituting this into the ratio yields:

$$HM/AB = 1/3 \Rightarrow AB = 3 \times HM$$

- **Proving $HA^2 = AB^2 - HM^2$:**

Since H lies on the line segment $[AC]$ and $(MH) \perp (AC)$, the triangle AHM is a right-angled triangle at H . Applying the Pythagorean Theorem in $\triangle AHM$:

$$HA^2 + HM^2 = AM^2$$

Because $AMNB$ is a rhombus, all its sides are equal in length, meaning $AM = AB$. Substituting AB for AM gives:

$$HA^2 + HM^2 = AB^2 \Rightarrow HA^2 = AB^2 - HM^2.$$

2) Direct Plane Similitude S

a) Precise the angle α of S and prove that its ratio $K = \sqrt{2}/4$.

- **Angle α :**

The direct similitude S has center H and transforms A into M , ($S(A) = M$). The angle α is defined by the directed vector angle: $\alpha = (\overrightarrow{HA}; \overrightarrow{HM})[\text{mod } 2\pi]$

Since $\triangle AHM$ is right-angled at H , the lines are perpendicular. Following the clockwise (negative) orientation from $\overrightarrow{HA}$ to $\overrightarrow{HM}$ based on the given geometric setup $(\overrightarrow{AC}; \overrightarrow{AB}) = \pi/2$, we find:

$$\alpha = -\pi/2 [\text{mod } 2\pi]$$

- **Ratio K :**

The ratio is defined as $K = HM/HA$. From part 1) c, we know that $AB = 3HM$, so $AB^2 = 9HM^2$. Substituting this into the Pythagorean identity:

$$HA^2 = AB^2 - HM^2 = 9HM^2 - HM^2 = 8HM^2$$

Taking the square root on both sides:

$$HA = \sqrt{8} HM = 2\sqrt{2} HM$$

Now, calculating the scaling ratio K :

$$K = HM/(2\sqrt{2}HM) = 1/2\sqrt{2} = \sqrt{2}/4$$

b) Determine the images of the two straight lines (AO) and (MH) under S . Deduce $S(N)$.

- **Image of (AO) : _under S ?**

We have $S: A \rightarrow M$, then the image of (AO) under S is a straight line passing through M and perpendicular to (AO) . Thus, $S(AO) = (MO)$.

- **Image of (MH) under S ?**

We have $S: H \rightarrow H$, then the image of (MH) under S is a straight line passing through H and perpendicular to (MH) . Thus, $S((MH)) = (HC)$.

- We have $S: \begin{cases} AO \rightarrow MO \\ MH \rightarrow HC \end{cases}$ and $(AO) \cap MH = \{N\}$ and $(MO) \cap HC = \{C\}$, then $S(N) = C$.

3) Midpoints and Tangency

Show that $S(O) = O'$: O is the midpoint of $[AN]$ and $S: \begin{cases} A \rightarrow M \\ N \rightarrow C \end{cases}$ then, $S(O) = O'$ which is the midpoint of $[MN]$. the midpoint O directly maps to the corresponding midpoint O' .

Deduce Tangency: $S: \begin{cases} O \rightarrow O' \\ H \rightarrow H \end{cases}$ and $\alpha = -\pi/2$, then (HO) is perpendicular to (HO')

H belongs to (C_2) and HO' is a radius of (C_2) , **Therefore, the line (HO) is tangent to (C_2) at H .**

4) Image of a Circle and Tangency

- Similitudes perfectly preserve geometric properties such as tangency and angles.
- From Part 3, we established that the straight line (HO) is tangent to the circle (C_2) at point H .
- Let's look at the images of these elements under S :

- The image of the circle (C_2) is (C'_2) .
- The image of the point of tangency H is $S(H) = H$.
- The image of the line (HO) is the line (HO') (since $S(H) = H$ and $S(O) = O'$).
- Since the transformation preserves the contact property, the image line (HO') must be tangent to the image circle (C'_2) at the image point H .
- Thus, the line $(O'H)$ is tangent to the circle (C'_2) at H .

5) Complex form of a similitude

a) S is a direct plane similitude of ratio $\sqrt{2}/4$ and angle $-\pi/2$, its complex form is given as :

$$z' = \frac{\sqrt{2}}{4} e^{-i\frac{\pi}{2}} z + b, O(0,0) \rightarrow O'(3,0). z_{O'} = \frac{\sqrt{2}}{4} e^{-i\frac{\pi}{2}} z_O + b ; \text{ leads to } b = 3$$

$$\text{therefore, } z' = -\frac{\sqrt{2}}{4} iz + 3.$$

$$b) z_H = \frac{b}{1-a} = \frac{3}{1 + \frac{\sqrt{2}}{4}i} = \frac{3}{1 + \frac{\sqrt{2}}{4}i} \cdot \frac{1 - \frac{\sqrt{2}}{4}i}{1 - \frac{\sqrt{2}}{4}i} = \frac{3(1 - \frac{\sqrt{2}}{4}i)}{1 - \frac{2}{4}} = \frac{3(1 - \frac{\sqrt{2}}{4}i)}{\frac{1}{2}} = 6(1 - \frac{\sqrt{2}}{4}i) = 6 - \frac{3\sqrt{2}}{2}i \rightarrow H(\frac{6}{3}; -\frac{2\sqrt{2}}{3})$$

$$c) z_H = \frac{z_M + z_D}{2} \rightarrow z_D = 2z_H - z_M = \frac{16}{3} - \frac{4\sqrt{2}}{3}i - 2 = \frac{10}{3} - \frac{4\sqrt{2}}{3}i$$

$$ID = |z_D - z_I| = \left| \frac{10}{3} - \frac{4\sqrt{2}}{3}i - 1 \right| = \left| \frac{7}{3} - \frac{4\sqrt{2}}{3}i \right| = \sqrt{\left(\frac{7}{3}\right)^2 + \left(\frac{4\sqrt{2}}{3}\right)^2} = \sqrt{9} = 3 = \text{radius of } (C_1)$$

Hence $D \in (C_1)$.

Question 5:

1) Verification of the Fundamental Algebraic Relation

We want to show that for all $z \neq i$, $(z - i)(z' - i) = 1$.

Substituting the explicit expression for z' into the left-hand side:

$$(z - i)(z' - i) = (z - i) \times [(iz + 2)/(z - i) - i]$$

By finding a common denominator within the brackets, we get:

$$= (z - i) \times [(iz + 2 - i(z - i)) / (z - i)]$$

Since $z \neq i$, the term $(z - i)$ non-zero and cancels out cleanly:

$$(z - i)(z' - i) = iz + 2 - i(z - i)$$

$$= iz + 2 - iz + i^2$$

$$= 2 + (-1) = 1$$

Hence, the relation holds true: $(z - i)(z' - i) = 1$.

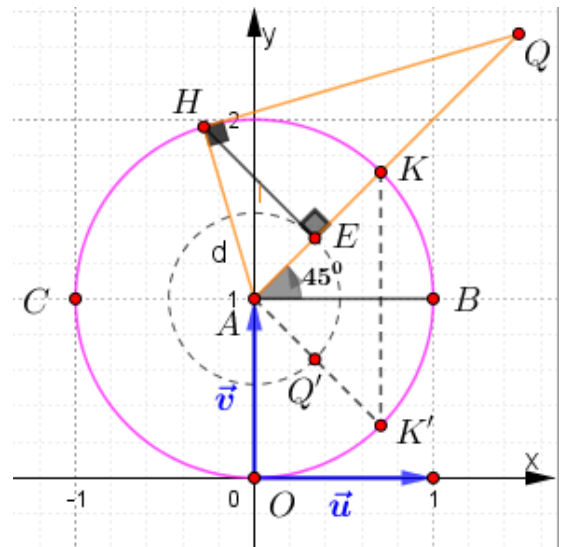
2) Deduction of the Non-Inclusivity Condition ($z' \neq i$)

From the identity proven in part (1), we have $(z - i)(z' - i) = 1$.

Let us perform a proof by contradiction. Suppose there exists a valid point $z \neq i$ such that its image satisfies

$z' = i$. This would immediately imply that:

$$(z' - i) = 0$$



Substituting this into our primary relation yields:

$$(z - i) \times 0 = 1 \Rightarrow 0 = 1.$$

This is an absolute mathematical contradiction. Therefore, our original assumption must be false.

Consequently, for all $z \neq i$, it is structurally impossible for the image to equal i , meaning $z' \neq i$.

3) Identification of Invariant Points ($M \equiv M'$)

To find the invariant or fixed points of this transformation, we solve the equation $M \equiv M'$, which translates to setting their complex affixes equal: $z = z'$.

$$z = (iz + 2)/(z - i)$$

Cross-multiplying by $(z - i)$ since $z \neq i$ gives:

$$z(z - i) = iz + 2 \Leftrightarrow$$

$$z^2 - iz = iz + 2 \Leftrightarrow$$

$$z^2 - 2iz - 2 = 0; (z - i)^2 - 1^2 = 0; (z - i - 1)(z - i + 1) = 0 ; \text{ leads to } z = 1 + i \text{ or } z = -1 + i$$

Conclusion: The set of points M satisfying $M \equiv M'$ consists of exactly two elements: $\{B, C\}$.

4) Geometric Analysis of Collinearity

a) Proving Collinearity Condition

Three points A, M , and M' are collinear if and only if the vectors $\overrightarrow{AM}$ and $\overrightarrow{AM'}$ are linearly dependent. In the complex plane, this requires the ratio of their corresponding complex vectors to be a purely real number:

$$(z_{M'} - z_A) / (z_M - z_A) = (z' - i)/(z - i) \in \mathbb{R}$$

From the fundamental relation in part (1), we know that $(z' - i) = 1/(z - i)$. Substituting this into our ratio gives:

$$(z_{M'} - z_A) / (z_M - z_A) = [1/(z - i)] / (z - i) = 1/(z - i)^2$$

For any non-zero fraction $1/W$ to be a real number, its denominator W must also be real. Therefore,

$$1/(z - i)^2 \in \mathbb{R} \text{ if and only if } (z - i)^2 \in \mathbb{R}.$$

Thus, points A, M , and M' are collinear if and only if $(z - i)^2$ is a real number.

b) Deducing the Locus of M

Let us denote the complex number $(z - i)$ in cartesian form as $X + iY$, where X and Y are real numbers.

Geometrically, $X = x$ and $Y = y - 1$, representing the translation of coordinates relative to the point $A(0,1)$.

Expanding the square :

$$(z - i)^2 = (X + iY)^2 = X^2 - Y^2 + 2iXY$$

For this expression to be strictly real, its imaginary component must vanish completely:

$$\text{Im}[(z - i)^2] = 0 \Rightarrow 2XY = 0 \Rightarrow X = 0 \text{ or } Y = 0$$

Let us evaluate these two geometric branches:

- Branch 1 ($X = 0$): This means $x = 0$, which corresponds to the y –axis (the imaginary axis).
- Branch 2 ($Y = 0$): This means $y - 1 = 0$, which simplifies to the horizontal line $y = 1$ passing through point A .

Since the transformation is explicitly undefined for $z = i$, point A must be extracted from the final locus set.

Conclusion: The geometric locus of M is the union of the imaginary axis ($x = 0$) and the horizontal line ($y = 1$), excluding the point A .

Second way: A, M , and $M' \Leftrightarrow (z - i)^2 \in \mathbb{R}$

$$\Leftrightarrow 2\arg(z - i) = k\pi, k \in \mathbb{Z} \text{ because } z - i \neq 0$$

$$\Leftrightarrow (\vec{u}; \vec{AM}) = k\pi/2, k \in \mathbb{Z}$$

Conclusion: The geometric locus of M is the union of the two straight lines (OA) and (AB) excluding the point A .

5) Metric and Angular Modulus Interpretations

Starting with the fundamental relationship $(z - i)(z' - i) = 1$, we can apply core complex function operations:

- Modulus Rule: Taking the modulus on both sides yields:

$$|(z - i)(z' - i)| = |1|$$

$$|z - i| \times |z' - i| = 1$$

Since $|z - i|$ represents the euclidean distance AM , and $|z' - i|$ represents AM' :

$$AM \times AM' = 1$$

- Argument Rule: Taking the principal argument on both sides yields:

$$\arg[(z - i)(z' - i)] = \arg(1)[2\pi]$$

$$\arg(z - i) + \arg(z' - i) = 0 [2\pi]$$

$$\arg(z' - i) = -\arg(z - i) [2\pi]$$

Notice that the vector AB is a horizontal unit vector extending from $A(i)$ to $B(1 + i)$, which points precisely along the positive real direction. Therefore, $\arg(z - i)$ represents the directed angle $(\vec{AB}; \vec{AM})$, and $\arg(z' - i)$ represents $(\vec{AB}; \vec{AM}')$. Substitution these gives :

$$(\vec{AB}; \vec{AM}') = -(\vec{AB}; \vec{AM})[2\pi].$$

Utilizing directed angle rotation properties, we know that $-(\vec{AB}; \vec{AM}) = (\vec{AM}; \vec{AB})$. Rearranging the components results in the targeted profile:

$$(\vec{AB}; \vec{AM}) = (\vec{AM}'; \vec{AB})[2\pi].$$

6) Mapping Characterizations for Point K

a) Metric and Angle Measurements for Image K'

We are given that point K has the affix $z_K = i + e^{i\pi/4}$. Rewriting this gives $z_K - i = e^{i\pi/4}$.

- Distance $AK = |z_K - i| = |e^{i\pi/4}| = 1$. This proves K is situated exactly on circle (C) .
- Angle $(\overrightarrow{AB}; \overrightarrow{AK}) = \arg(z_K - i) = \pi/4$ (or 45°).

Applying the specific mapping transformations deduced in part (5) directly to point K :

- Distance calculation: $AK \times AK' = 1 \Rightarrow 1 \times AK' = 1 \Rightarrow AK' = 1$.
- Angular calculation: $(\overrightarrow{AB}; \overrightarrow{AK'}) = -(\overrightarrow{AB}; \overrightarrow{AK}) = -\pi/4$ (or -45°).

Summary: The distance $AK' = 1$, and the directed angle measure $(\overrightarrow{AB}; \overrightarrow{AK'}) = -\pi/4$.

b) Graphical Plotting Guide for K'

We have $AK = AK'$ and $(\overrightarrow{AB}; \overrightarrow{AK'}) = -(\overrightarrow{AB}; \overrightarrow{AK})$ then, K' is the orthogonal symmetry of K with respect to the straight line (AB) .

c) Locus Transformation of the Semi-Straight Line

When M traces the semi-straight line $[AK)$ excluding point A , the angle is permanently fixed at $(\overrightarrow{AB}; \overrightarrow{AM}) = \pi/4$.

Consequently, its image point M' must maintain a fixed angular path satisfying $(\overrightarrow{AM'}; \overrightarrow{AB}) = -\pi/4$.

As M moves outward from point A towards infinity (meaning the length AM spans from 0 to ∞), the inverse relationship dictating the distance requires $AM' = 1/AM$ to span from ∞ down towards 0 (approaching but never reaching A).

Conclusion: The set of image points M' traces the semi-straight line $[AK')$ pointing at an angle of -45° , excluding the boundary point A .

7) Geometric Projection and Inversion Properties

a) Metric Inversion Proof: $AQ \times AE = 1$.

Based on the provided geometric configuration, points K and Q share an argument of $\pi/4$ relative to A . Thus, A, K, E , and Q are strictly collinear along the same ray line.

We are given that triangle AHQ is a right-angled triangle at H ($\angle AHQ = 90^\circ$), and E is the orthogonal projection of vertex H onto the hypotenuse ray line AQ . This setup allows us to apply the right-triangle Euclidean geometric projection theorem:

$$AH^2 = AE \times AQ$$

Because H is defined as an element on the circle (C) which has a fixed radius of 1 centered at A , the segment length AH equals 1. Substituting this radius value gives:

$$1^2 = AE \times AQ \Rightarrow AQ \times AE = 1.$$

Hence, the metric inversion property is proven: $AQ \times AE = 1$.

b) Precise Geometrical Construction of Q'

We have, $AE \times AQ = 1$ and $AQ \times AQ' = 1$, then $AE = AQ'$ this leads to: Q' belongs to the circle (S) of center A and radius AE .

On the other hand, Q' belongs to the semi line $[AK')$ except A

We can conclude that $\{Q'\} = (S) \cap [AK') - \{A\}$